Lesson Plan

Teacher Name: Ms. Lubna Kakkengal and Mr. Dilan Devendra	**Class/Section:** Grade 6
Subject: TCD: My Maker Mark	**Date: Week 10 (2 Nov to 6 Nov)**
Class Description: Grade 6 TCD class, My Maker Mark Session 2 of 3 (2 hours). Students prepare their maker mark as a laser file with separate cut and engrave layers, design the outline of a keychain or name plaque, and watch their piece being laser cut by the teacher.	
TA/LSA Role (if applicable):	

Differentiated Learning Groups (Data)			
Desert Fox	**Camel**	**Oryx**	**Falcon**

Lesson Overview		
Lesson Topic	Laser Cutting My Maker Mark	
Standard(s)	Math: 6.G.A (area and dimensions of shapes), 6.RP (scale); Technology: 2D CAD to fabrication workflow; Science: material properties	
Learning Objective(s) (WALT): We are learning to prepare a 2D design for the laser cutter and to understand how the laser cuts and engraves.		**Success Criteria (WILF):** I can explain the difference between cutting and engraving. I can set up a file with a cut outline and an engrave area. I can choose a suitable size and material for my piece. I can follow the laser cutter safety rules.
Literacy Strategies	Laser vocabulary cards (vector, raster, cut, engrave, kerf, power, speed); frame: The laser will cut ... and engrave ...	
Assessment	Laser-ready file checked by the teacher; observation of safety routines; live quiz on the TCD website.	
Targeted Vocabulary	laser cutter, vector, raster, cut, engrave, kerf, plywood, acrylic, power, speed	
Skills	Technical preparation, measuring, safety awareness, attention to detail	
National Identity	Link to laser-cut mashrabiya patterns in modern UAE	

Integration	architecture, such as the shading screens on buildings in Abu Dhabi and Dubai.
AI/Digital Learning	Laser software on the teacher's computer (for example LightBurn or the laser's own software); Canva files from Week 9; hook video; live quiz on the TCD website. Presentation: Grade6_Week10_Presentation.pptx. Worksheet: Grade6_Week10_Worksheet.

Lesson Structure	
Starter (Hook) Engage and Ask (15 minutes)	Warm-up: Cut or Engrave? (slide 4). Show the hook video and ask the hook question (slide 2). Share the learning goals (slide 3) and key words (slide 5).
I Do (25 minutes)	Explain how the laser works (slide 6). Model setting cut and engrave layers on a sample file (slide 7). Explain size, material and kerf (slide 8). Go through laser safety (slides 9 and 11).
We Do (20 minutes)	Students import their maker mark and draw the outline with teacher support (slide 10). Partners check each other's files against the checklist.
You Do (45 minutes)	Students finalise their laser file and get the teacher's check (slide 12). The teacher runs the laser in batches while students observe from behind the safety line and record settings (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: **Beginning / On / Above**	**Beginning: Use the ready-made circle keychain outline and place your maker mark inside.** **On: Design your own outline shape with a hole and correct layers.** **Above: Design a plaque with your maker mark and a short engraved motto, and calculate how many pieces fit on one sheet.**
Closure/ Reflection (15 minutes)	Students compare finished pieces with their file. What changed? Exit ticket (slide 15). Pieces are kept for finishing next week.

Inclusive Planning
Students receiving additional inclusive support (IEP or ILPs) or Gifted and Talented: EAL: laser vocabulary cards with pictures. SEN: ready-made outline template and partner support. G&T: plaque with motto and material efficiency calculation.
Inclusion / EAL Support

☑ Visual Supports
☑ Chunked Instructions
☐ Reduced Language Load
☑ Model / Worked Example
☑ Pre-teach Vocabulary
☐ TA/LSA Support
☐ Extra Processing Time
☐ Text Read by TA/LSA
☐ Flexible Grouping
☐ Assistive Technology
☐ Movement Break
☐ Scaffold / Sentence Starters
☐ Other:

G&T / Extension

☑ Extension / Challenge Task
☐ Higher-order Questioning
☑ Depth & Complexity Task
☐ Independent Inquiry
☐ Leadership / Peer Coaching
☑ Real-World Application
☐ Curriculum Compacting
☐ Above-grade-level Texts
☐ Student-led / Passion Project
☐ Open-Ended Task
☐ Other:

Assessment Adaptations

☑ Oral Response
☐ Reduced Questions
☐ Scaffolded Success Criteria
☐ Extra Time
☐ Alternative Outcome Format
☑ Teacher Conferencing/Live Feedback
☐ Quiet / Separate Space
☐ Other: